

Week 4

Notation

$$A = [a_{ij}]$$

$i = 1, \dots, m \rightarrow$ rows

$j = 1, \dots, n \rightarrow$ column

$$\dim(A) = m \times n$$

~~Ex~~

3, 4



3+4

↑
add

3 · 5

↑
multiplication

$$A_2 = \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix} \quad B = [2]$$

you cannot add

$$\dim(A) = 2 \times 2 \quad \dim(B) = 1 \times 1$$

~~A + B~~

Matrix addition

$$C = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$$

$$C + A = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

you can always do

$$2. \underbrace{\begin{bmatrix} 1 & 1 & 3 \\ 2 & 3 & 4 \end{bmatrix}}_{\dim(A) = 2 \times 3} = \underbrace{\begin{bmatrix} 2 & 2 & 6 \\ 4 & 6 & 8 \end{bmatrix}}_{2 \cdot A}$$

$\Rightarrow$ think about numbers: 0

$$a + 0 = a$$

$\uparrow$ zero

$$A = B$$

$$\dim(A) = \dim(B)$$

$$a_{ij} = b_{ij}$$

$\Rightarrow$ Q: "Zeros"?

infinitely many different

Zeros

$$\begin{matrix} & 0_{1 \times 1} & 0_{2 \times 2} & 0_{3 \times 2} \end{matrix}$$
$$\begin{bmatrix} 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

more complicated operations

X $\odot$ $\leftarrow$ does not
exist in matrix

$\hookrightarrow$ Nothing like you've seen before

$$\hookrightarrow 3.4 = 12$$

$$A = \begin{bmatrix} 1 & -3 & -1 \\ 2 & 4 & 5 \end{bmatrix} \quad \dim(A) = 2 \times 3$$

$$B = \begin{bmatrix} 1 & 0 & 2 & 5 \\ -1 & 1 & 3 & 1 \\ 2 & -2 & -1 & -3 \end{bmatrix} \quad \dim(B) = 3 \times 4$$

must be same

$$\dim(A \cdot B) = (2 \times 3) \cdot (3 \times 4) = (2 \times 4)$$

Not same

$$\dim(B \cdot A) = (3 \times 4) \cdot (2 \times 3) = \text{DNE} \quad \dim(A \cdot B)$$

Cannot multiply

$$\begin{bmatrix} 1 & -3 & -1 \\ 2 & 4 & 5 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 2 & 5 \\ -1 & 1 & 3 & 1 \\ 2 & -2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -6 \\ 8 & -6 & 11 \end{bmatrix} \quad (2 \times 4)$$

$$\begin{aligned}
 (1) \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \begin{bmatrix} -3 \\ 4 \end{bmatrix} + (2) \begin{bmatrix} -1 \\ 5 \end{bmatrix} &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -4 \end{bmatrix} + \begin{bmatrix} -2 \\ 10 \end{bmatrix} \\
 &= \begin{bmatrix} 1 + 3 - 2 \\ 2 - 4 + 10 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 (1) \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} -3 \\ 4 \end{bmatrix} + (-2) \begin{bmatrix} -1 \\ 5 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -3 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -10 \end{bmatrix} \\
 &= \begin{bmatrix} -1 \\ -6 \end{bmatrix}
 \end{aligned}$$

Hes not doing column 4

$$2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \cdot \begin{bmatrix} -3 \\ 4 \end{bmatrix} + (-1) \begin{bmatrix} -1 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} -9 \\ 12 \end{bmatrix} + \begin{bmatrix} 1 \\ -5 \end{bmatrix} = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$$

$$A = \begin{bmatrix} x \\ x \\ x \\ x \end{bmatrix} + \begin{bmatrix} x & x & x & x \end{bmatrix} = \begin{bmatrix} x & x & x & x \\ x & x & x & x \\ x & x & x & x \\ x & x & x & x \end{bmatrix}$$

able to write SLE in matrix form

$$\begin{cases} 3x - 2y + z = 7 \\ 4x - 6y + 2z = 3 \end{cases} \rightarrow \begin{bmatrix} 3 & -2 & 1 & 7 \\ 4 & -6 & 2 & 3 \end{bmatrix}$$

augmented
matrix

$$A = \begin{bmatrix} 3 & -2 & 1 \\ 4 & -6 & 2 \end{bmatrix}$$

$$\dim(A) = 2 \times 3$$

$$\bar{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \dim(\bar{X}) = 3 \times 1$$

$$\bar{b} = \begin{bmatrix} 7 \\ 3 \end{bmatrix} \quad \dim(\bar{b}) = 2 \times 1$$

$$A \cdot \bar{X} = \bar{b} \Leftrightarrow \begin{cases} 3x - 2y + z = 7 \\ 4x - 6y + 2z = 3 \end{cases}$$

Note, in matrix multiplication $A \cdot \bar{X}$

$$(2 \times 3) \cdot (3 \times 1) = 2 \times 1 = \dim(\bar{b})$$

$$\begin{matrix} \begin{bmatrix} 3 & -2 & 1 \\ 4 & -6 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \\ \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \end{bmatrix} \\ \begin{matrix} (2 \times 3) & (3 \times 1) & = (2 \times 1) \end{matrix} \end{matrix}$$

$$x \begin{bmatrix} 3 \\ 4 \end{bmatrix} + y \begin{bmatrix} -2 \\ -6 \end{bmatrix} + z \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3x - 2y + z \\ 4x - 6y + 2z \end{bmatrix}$$

$$\left. \begin{array}{l} A + B \\ \lambda \cdot A \\ A \cdot B \end{array} \right\} \text{basic matrix}$$

"transpose"

$$\dim A = m \times n$$

rows

columns

$$\dim(A^T) = n \times m$$

↑
rearranges matrix

$$A \xrightarrow{T} A^T$$

↑ transpose of A

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -5 & 6 \end{bmatrix} \xrightarrow{T} A^T \begin{bmatrix} 1 & 3 \\ 2 & -5 \\ -1 & 6 \end{bmatrix} \rightarrow (A^T)^T \begin{bmatrix} 1 & 2 & -1 \\ 3 & -5 & 6 \end{bmatrix}$$

must be the same size

$$\swarrow \searrow$$
$$A + B = B + A$$

$$(A+B)+C = A+(B+C)$$

$$A \cdot (B+C) = A \cdot B + A \cdot C$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

$$\dim(A) = \dim(B)$$

$$\checkmark \underline{A \cdot B} \neq B \cdot A \checkmark$$

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & -2 \\ 3 & 0 \end{bmatrix}$$

$$A \cdot B = \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 2 & -2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 8 & -2 \\ 13 & 2 \end{bmatrix}$$

$$B \cdot A = \begin{bmatrix} 2 & -2 \\ 3 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 4 & -6 \\ 3 & 6 \end{bmatrix}$$

$\neq$

$$O_{m \times n} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 \end{bmatrix}}_n \leftarrow \text{zero matrix}$$

↳ Identity matrix must be a $\square_n^n$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{\text{not important}} A_{n \times m} \xrightarrow{I} n \times m \quad \text{diagonal doesn't change}$$

main diagonal

$$I_n = [a_{ij}], \quad a_{ij} = \begin{cases} a_{ii} = 1, & i = 1 \dots n \\ a_{ij} = 0, & i \neq j \end{cases}$$

for any $n \times n$ matrix

$$\hookrightarrow \underbrace{A \cdot I_n = A,}$$

$$\underbrace{I_n \cdot A = A}$$

Claim

$$A \xrightarrow{ERO} A_{REF} \xrightarrow{ERO} A_{RREF}$$

$$\begin{array}{c} \text{A} \\ \text{---} \end{array} \xrightarrow{\text{nxn}} \begin{array}{c} A_{RREF} \\ \text{---} \end{array} = \begin{array}{c} \textcircled{1} \\ \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \\ \textcircled{2} \\ \left[\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

Every Number has inverse except zero

A ∞ Matrix doesn't have inverse

If A is not a $\square$ $\xrightarrow{\text{stop}}$ No A^{-1}

inverse to A
↙

$$A \cdot A^{-1} = I_n \quad \text{and} \quad A^{-1} \cdot A = I_n$$

A square non-invertible matrix is called singular

↳ $n \times n$

• $n = 1 \longrightarrow \dim(A) = 1$

• $n = 2$

• $n = \text{any} \geq 3$

$A = [a] \xrightarrow{\text{trivial}} A^{-1} \left[\frac{1}{a} \right]$
 $a \neq 0$

$n = 2$

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

When A is
invertable?
(A^{-1} exists)

↳ determinant

$\det(A) = a \cdot d - b \cdot c$

memorize

main diagonal

off diagonal

$= 0$ Full stop

$\neq 0 \rightarrow A^{-1}$ exists

$A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \frac{1}{\underbrace{ad - bc}_{\det(A)}} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \frac{1}{\det(A)}$
??
memorize

Practice

$$A = \begin{bmatrix} 5 & 0 \\ 6 & 3 \end{bmatrix}$$

↓

$$\det(A) = 5 \cdot 3 - 6 \cdot 0 = 15 - 0 = 15 \neq 0$$

$$= \begin{bmatrix} \frac{3}{15} & \frac{0}{15} \\ -\frac{6}{15} & \frac{5}{15} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} & 0 \\ -\frac{2}{5} & \frac{1}{3} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{1}{5} & 0 \\ -\frac{2}{5} & \frac{1}{3} \end{bmatrix}$$

check with

$$A \cdot A^{-1} = I_n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Learn elementary matrix
by myself. will be on test

⇒ Example:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Follow the algorithm.

$$\left[A \mid I_3 \right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow r_3 \rightarrow r_3 - r_1 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array} \right]$$

$$\rightarrow r_3 \rightarrow r_3 - r_2 \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array} \right] \rightarrow r_3 \rightarrow -\frac{1}{2} \cdot r_3 \rightarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right] \rightarrow \begin{array}{l} r_1 \rightarrow r_1 - r_3 \\ r_2 \rightarrow r_2 - r_3 \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \end{array} \right] = \left[I_3 \mid A^{-1} \right]$$

⇒